CodeCraft Lesson Plan

10 sessions, 1 hour 30 minutes each
Audience: secondary students (roughly ม.4-ม.6)
Approach: one project per session, moving from visual UI building to logic, navigation, plugins, and sharing

Course Goals

By the end of the course, students should be able to:

- build mobile-style screens in Design mode
- connect elements with Code mode
- use inputs, labels, buttons, lists, and navigation
- understand page.add(...), onClick(...), and page.goTo(...)
- work with simple state using .value, .text, .content
- preview, test, and share projects with PWA links

Session 1: My First Screen

Project

Profile Card App

Learning Focus

- understand the editor layout
- add elements to the canvas
- edit properties
- use Heading, Label, Image, Card, Spacer

Outcome

Students build a simple profile screen with picture, name, class, and short bio.

Suggested Flow

- 15 min: intro to CodeCraft workspace
- 20 min: teacher demo of adding and editing elements
- 40 min: students build the profile card
- 15 min: share and reflect

Homework / Extension

Change the screen into a favorite artist, athlete, or game character card.

Session 2: Interactive Buttons

Project

Mood Button App

Learning Focus

- add Button, Label, Heading
- enter Code mode

- use page.add(...)
- write first onClick(...)
- update text using .content

Outcome

Students make buttons like Happy, Sad, Excited, Sleepy and update a label when clicked.

Sample Challenge

When clicking each button, show a different message in the result label.

Session 3: Simple Calculator

Project

Two-Number Calculator

Learning Focus

- use Input, Label, Button
- read .value
- convert text to numbers
- show result using .content

Outcome

Students build addition, subtraction, multiplication, and division buttons.

Suggested Stretch

Show an error message when dividing by zero.

Session 4: Quiz App

Project

One-Page Quiz Checker

Learning Focus

- use Checkbox, Radio, Label, Button
- compare answers
- show score and feedback

Outcome

Students create a 3-5 question quiz with a Check Answer button.

Suggested Stretch

Use color or badges to show correct / incorrect results.

Session 5: Daily Planner

Project

Mini Planner App

Learning Focus

- use DatePicker, Textarea, Label, Separator, Spacer
- structure content clearly on screen
- collect user input and display summary

Outcome

Students make a planner screen where they choose a date and write a short task plan.

Suggested Stretch

Add a button that displays: “Your plan for [date] is ready.”

Session 6: Multi-Page Story App

Project

Choose Your Adventure Story

Learning Focus

- create multiple pages
- use page.goTo(...)
- connect buttons to different story endings

Outcome

Students build a branching story with at least 3 pages.

Suggested Stretch

Make one good ending and one bad ending.

Session 7: Shop Menu App

Project

Snack Shop Ordering Screen

Learning Focus

- use List, Card, Segment, Badge, Button
- organize app content visually
- build selection flow with logic

Outcome

Students create a small snack menu and show the chosen item in a result label.

Suggested Stretch

Use tabs or segments such as Drinks, Snacks, Desserts.

Session 8: Smart Device Controller

Project

Home Control Panel

Learning Focus

- use Toggle, Range, Label, Icon
- simulate device state changes
- update labels based on controls

Outcome

Students build a control panel for fan speed, room lights, or air conditioner settings.

Suggested Stretch

Show text like “Light is ON” or “Fan speed = 80”.

Session 9: Camera and Media App

Project

Photo Booth Starter

Learning Focus

- use Camera, Button, Label
- understand plugin-like behavior
- open and close camera with code
- discuss browser permissions and testing conditions

Outcome

Students build a simple app where clicking a button opens the camera view.

Example Code Goal

```
    var camera1 = new Camera('page1_camera1');
    var button1 = new Button('page1_button1');

    page.add(camera1);
    page.add(button1);

    button1.onClick(() => {
      camera1.open();
    });
```

Suggested Stretch

Add a second button to close the camera.

Session 10: Final Project and Share Day

Project

Student Choice App

Learning Focus

- plan app flow
- combine design + code
- debug with console
- run preview and share with PWA link

Outcome

Students create and present one complete mini app using skills from previous sessions.

Suggested Project Choices

- study timer
- mood journal
- mini food order app
- class voting app
- travel checklist
- simple health tracker

Presentation Goal

Each student or pair should show:

- what the app does
- what elements they used
- one code part they understand well
- one challenge they solved

Recommended Session Structure

Use this same rhythm each week:

- 10 min: review last session
- 15 min: teacher demo
- 45 min: student build time
- 10 min: debugging / help checkpoints
- 10 min: showcase or reflection

Skills Map By Session

- Session 1: layout and property editing
- Session 2: buttons and click events
- Session 3: input and output logic
- Session 4: conditions and checking answers
- Session 5: forms and structured content
- Session 6: multi-page navigation
- Session 7: organizing data visually
- Session 8: simulated controls and dynamic feedback

- Session 9: plugin thinking and browser permissions
- Session 10: independent app design and presentation

Assessment Ideas

- build completion
- screen layout quality
- correct use of elements
- correct use of page.add(...)
- correct use of onClick(...)
- correct use of .value, .content, or .text
- presentation clarity

Teacher Notes

- Keep Session 1-3 very guided.
- Let students personalize visuals early to keep motivation high.
- From Session 6 onward, allow more creative freedom.
- Pair programming works especially well for Session 9 and Session 10.
- Use the share link often so students feel they are building real apps for phones.